

Session 15: Post Class tests

1. You have been presented with estimated after-tax operating income and working capital requirements on a project that has a five-year life.

	1	2	3	4	5
After-tax Operating Income	\$100	\$120	\$135	\$145	\$150
Working Capital (Total)	\$20	\$24	\$27	\$29	\$30

The project requires an initial investment of \$150 million in new capacity that will depreciated straight line to a salvage value of \$30 million at the end of year five. Assuming that you will be able to recoup your entire working capital at the end of year 5, estimate the after-tax cash flows each year for the next 5 years.

	Now	1	2	3	4	5
Initial Investment						
After-tax Operating Income		\$100	\$120	\$135	\$145	\$150
+ Depreciation						
- Change in working capital						
+ Salvage value						
After-tax cash flow						

2. You are reviewing the cash flows estimated by an analyst for a ten-year project. The after-tax cash flow each year is expected to be \$25 million, but those cash flows are after an allocation of \$ 10 million in G&A expenses to this project each year. If only 25% of these G&A expenses are variable (and related to this project) and the tax rate is 40%, what is the correct incremental cash flow each year on this project?
- \$26.0 million
 - \$ 26.5 million
 - \$ 27.5 million
 - \$ 28.0 million
 - \$29.5 million
 - \$32.5 million
3. You are looking at project with a 10-year life. The initial investment is \$100 million and is depreciable straight-line to a salvage value of \$20 million and the after-tax operating income each year is \$12 million. There are no capital maintenance or working capital requirements. The cost of capital for the company taking the project is 8% but this project is in a riskier business, with a cost of capital of 12%. The marginal tax rate is 40%. What is the correct NPV of this project?
- \$7.68 million
 - \$1.36 million
 - \$13.00 million
 - \$19.44 million
 - \$43.47 million

4. You have been asked to analyze the net present value of building a toll road in Asia. You estimate that building the road will cost you \$50 million up front and that you will generate \$ 4 million in cash flows next year and that these cash flows will grow 10% a year for the following four years (Years 2-5). After year 5, you expect the cash flows to continue to grow at the inflation rate (2%). Assuming a cost of capital of 8%, what is the NPV of this project to you?
- a. \$19.04
 - b. \$31.96
 - c. \$35.65
 - d. \$36.98
 - e. None of the above
5. Now assume that you believe that you can double your cash flows for the next 5 years in the toll road investment described in problem 2, if you cut back on maintenance. If you do this, though, you will no longer be able to operate it as a toll road after year 5 and will have to sell it to the government for \$20 million. Assuming the cost of capital remains at 8%, what is the NPV of the project to you?
- a. -\$ 11.56 million
 - b. \$1.04 million
 - c. \$2.05 million
 - d. \$ 8.43 million
 - e. None of the above
6. Revere Inc. is a publicly traded company that manufactures kitchen utensils; it has 100 million shares outstanding, trading at \$15/share. It has decided to acquire Luzo Inc., a small competitor, for \$150 million. If Revere's stock price drops to \$14.50/share on the announcement, what is the value that the market is attaching to Luzo Inc.?
- a. \$50 million
 - b. \$100 million
 - c. \$150 million
 - d. \$200 million
 - e. None of the above

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- To compute the cash flows, you first have to compute the depreciation each year:
 $\text{Depreciation} = (150 - 30) / 5 = 24$. The change in working capital over the previous year also has to be computed. The salvage value will be the sum of the salvage value of the initial investment (\$30 million) and the working capital (\$30 million)

	Now	1	2	3	4	5
Initial Investment	-150					
After-tax Operating Income		\$100	\$120	\$135	\$145	\$150
+ Depreciation		24	24	24	24	24
- Change in working capital		20	4	3	2	1
+ Salvage value						60
After-tax cash flow	-150	104	140	156	167	233

- d. \$29.5 million.** Since these are after-tax cash flows, you have to add back the after-tax fixed cost portion of the G&A (75% of the allocated amount) to get to the corrected cash flow
 - Corrected cash flow = \$25 million + \$7.5 million (1-4) = \$29.5 million
- d. \$19.44 million.** The NPV should be computed using the higher cost of capital. To compute the correct NPV, we first compute the after-tax cash flow each year.
 - After-tax cash flow = After-tax Operating Income + Depreciation = \$12 + $(100 - 20) / 10 = \$20$ million
 - In year 10, you will have a salvage value of \$20 million
 - $\text{NPV} = -100 + 20 (\text{PV of annuity, 10 years, 12\%}) + 20 / 1.12^{10} = \19.44 m
- d. \$36.98 million.**

	Now	1	2	3	4	5
Investment	-\$50.00					
ATCF		\$4.00	\$4.40	\$4.84	\$5.32	\$5.86
Terminal value						\$99.56
PV @8%	-\$50.00	\$3.70	\$3.77	\$3.84	\$3.91	\$71.75
NPV	\$36.74					

Terminal value = $\$5.86 (1.02) / (.08 - .02) = \99.56 million

- c. \$2.05 million**

ATCF	-\$50.00	\$8.00	\$8.80	\$9.68	\$10.65	\$11.71
Salvage value						\$20.00
PV @8%	-\$50.00	\$7.41	\$7.54	\$7.68	\$7.83	\$21.58
NPV	\$2.05					

6. **b. \$100 million.** The equity value of the acquirer (Revere) dropped by \$50 million on the announcement of the acquisition. The market, therefore, thinks that Revere paid \$50 million too much, when it acquired Luzo for \$150 million.
- Market's estimate of Luzo's value = $\$150 - \$50 \text{ m} = \$100 \text{ million}$